

Chapter 5

Security: Windows and Linux System Security

By Dr. Samuel Asferaw
2025/2026



Contents

- **Windows System Security**
- **Linux System Security**

Windows Security

Basics

- **Passwords**

- Strong password should:

- Be long
 - Contain:
 - Letters
 - Punctuation
 - Symbols
 - Numbers

- Password Checker

- <http://www.microsoft.com/security/online-privacy/password-checker.aspx>

Password ...

- Password is known **ONLY** to **user**
- Storing/Managing Passwords
 - Password algorithms
- Password files

File System

- **NTFS** Security Advantages
 - Access Control List
 - Granular (Rough) structure
 - Supports server authentication
 - Ability to encrypt files and directories
 - Encrypted File System (EFS)

NTFS Access Control

PERMISSIONS

For files, directories, networked devices

APPLICABLE ON:

- Users
- Computers
- Networked devices
- Groups of:
 - Users
 - Computers
 - Networked devices

INHERITANCE

Surfing the Internet

- **Browsers**

- Firefox (FF) and
- Internet Explorer (IE)
- Others (Opera, Chrome, etc)

- **Browser Add-ons**

- NoScript
- FlashBlock

Environmental Threats

COMPONENT OBJECT MODELS

- Object Linking and Embedding (OLE)
 - Remote Procedure Call (RPC)
 - ActiveX

JAVA APPLET

External Threats

- **Browser related**
 - Hijacks (BHO)
- **Drive-by downloads**
 - WMF (2005), ActiveX, DCOM
- **Scams**
 - Confidence tricking
- **Phishing**
 - Fake Emails
- **ID Theft**
 - Data harvesting
- **Social Engineering**
 - Psychological manipulation
- **Targeted Malware**
 - Silent_banker

Human Error

- Emails
- Downloads
 - P2P
 - Underground sources
- System patches out of date
- Clicking without reading/fully understanding ('Nexters')

Privacy

- Indexing services
 - Google Desktop
- Social websites
 - Facebook
 - ID Threats
 - Facebook Applications
 - Source of infection
 - Data mining
- Search engines
 - Google
 - Scrapped google engine (Scroogle): providing Google search results in a **private setting, free of ads, tracking, and cookies**. But, it was closed in 2012
 - Ixquick: a search engine that is third- party certified not to record users' IP addresses or make a record of their searches
 - DuckDuckGo : is an independent Google alternative that also lets you search and browse the web, but we don't track your searches or browsing history

Operating System Security Software

- Anti Virus
 - Microsoft Security Essentials (MSE)
- Anti Spy/Malware
 - Windows Defender
- Tracking Software
 - Adeona
- Firewall
 - Windows Firewall
 - Sygate Personal Firewall

Common Sense

- Strong Password
- Latest updates
- Unprivileged user account
- Read **EVERYTHING** on screen
- Never disclose any login details
- **Never believe in something for nothing**
- Be Aware
- Better safe than sorry

Safe System

- **Internet Browsing**

- Tightened settings for Browsers
- Do not log in as Administrator
- Build up adequate layer of defence through application layer
- Real-Time anti-virus/spy/mal/ad ware

- **Electronic Mail**

- Set up for plain text only
- Be careful what you subscribe to
(mailing list harvesting)
- Spam filters
- Email address obfuscating: hides email addresses from spam bots on websites
- Structuring multiple email addresses for different purposes

- **System layer**

- Hidden files (double extensions)
- Start-up
- Active software protection

Linux Security

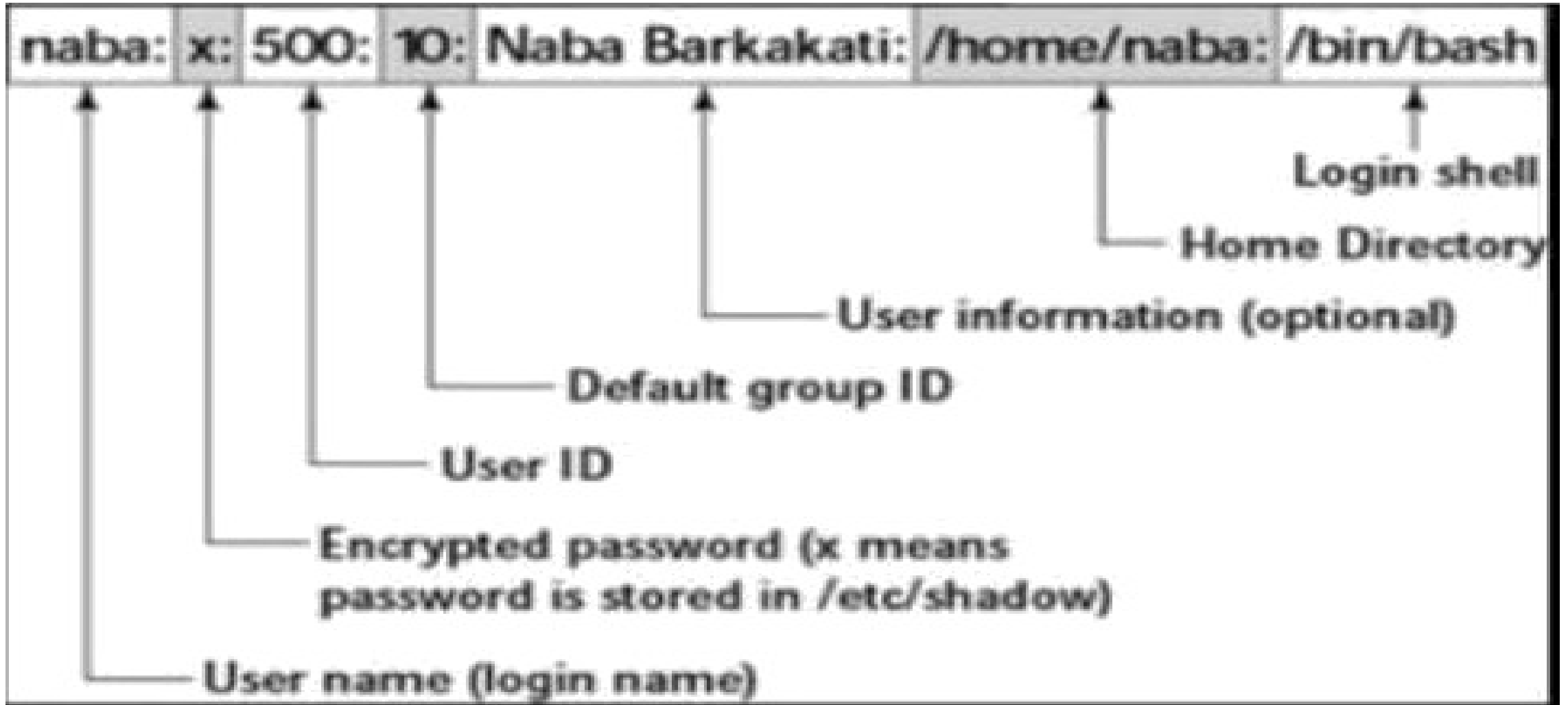
Introduction to Linux Security

- Being a **multi-user system** - Linux has a tremendous amount of security to offer; many being open source, which can be validated and modified to meet anyone's needs.
- Security mechanisms allow users to service their legitimate needs without compromising the server system security.
- Resource and file management maintain integrity
- User privileges are granted.
- Overall system stability and correctness do not suffer.

Authentication

- User enters username and password via login
- Passwords are hashed .
- Encryption cannot be reversed Stored in /etc/passwd or /etc/shadow
- Pluggable authentication modules (PAMs) Can reconfigure the system at run time to include enhanced authentication techniques
- Supports smart cards, Kerberos and voice authentication

Understanding the /etc/passwd



Cryptography

- Enables users to access several forms of encryption to protect their data
- Uses powerful algorithms such as DES, AES and MD5
- Kernel uses Cryptographic API to implement IPSec
- Enables users to create secure (encryptRd) file systems

- **Loopback device:**

Layer between the virtual file system and the existing file system
Can be used to encrypt and decrypt data transferred between processes and the underlying file system

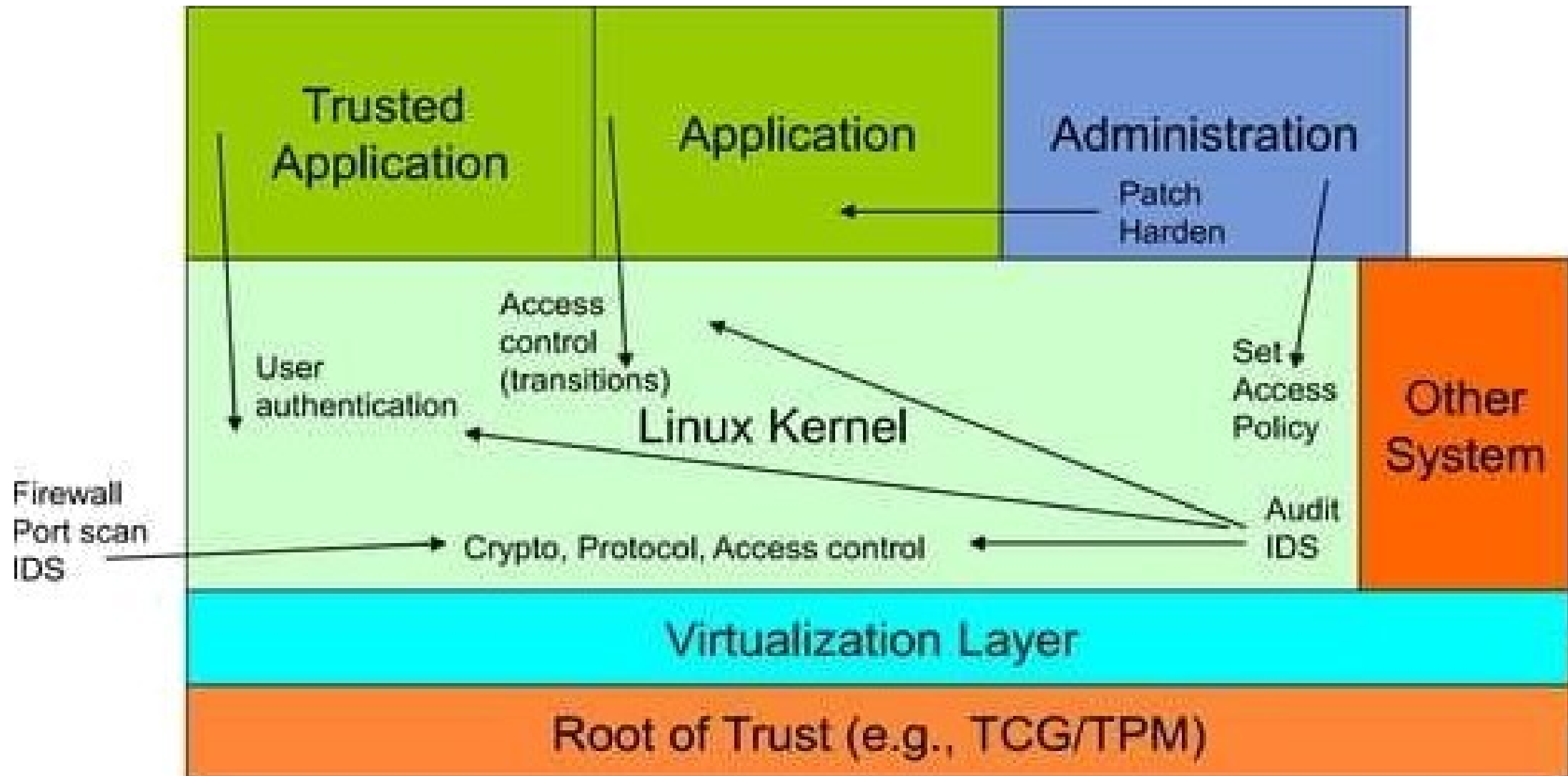
Administering Trusted Users and Hosts

- The **.rhosts** file exists in a user's home directory
- Specifies trusted hosts based on the users choice
- More headaches for administrator - loss of control
- **Solution** - disable or monitor contents.

Limiting User Access to System Resources

- In **/etc/pam.d/limits.conf**
- Limit processes per user.
- Limit memory usage.

System Security View



ACL/MAC Permissions

Access Control List

- Almost modern files systems, include ACLs to give unprivileged access to only certain users.
- Permissions are separated by owner, group and others.
- Permissions re displayed as rwx rwx rwx .

Mandatory Access Control

- More sophisticated form of permissions handling. This is more like application patching, it limits what permission each program is given.
- MAC programs include: AppArmor, SELinux, SEBSD, GrSecurity, Trusted Solaris and Trusted BSD.

Memory Access

- Each process has its own page table.
- All memory access via page table.
- Easy for OS to terminate process which references an invalid memory address.
- Access control information for page held in the page table entry (PTE).
- Prevents executable code from being overwritten
- Separates kernel code and user code.

Firewalls

- Firewalls are means of controlling what information is allowed into and out of your local network.
- Linux Firewalls are:
 - IPTables
 - SELinux (Security-Enhanced Linux) is a Linux kernel security module that provides mandatory access control (MAC), adding a crucial security layer beyond traditional file permissions
 - Scalable
 - Robust

Graphical Firewall Configuration Utilities

- Linux supports several graphical tools that can be used to set up a firewall
- Red Hat Linux includes the lokkit program that walks you through questions and establishes rules based on your security choices
- Red Hat Linux also includes the firewall-config program, which allows the set up of complex firewall rules

Conclusion

- Linux is a versatile OS.
- Security implementation in the OS is **spread throughout the system - memory management, file management, process management**, etc.
- **Therefore**, every aspect of security needs to be configured from scratch since default is not maximum security.

End of Course!



Thank you!